

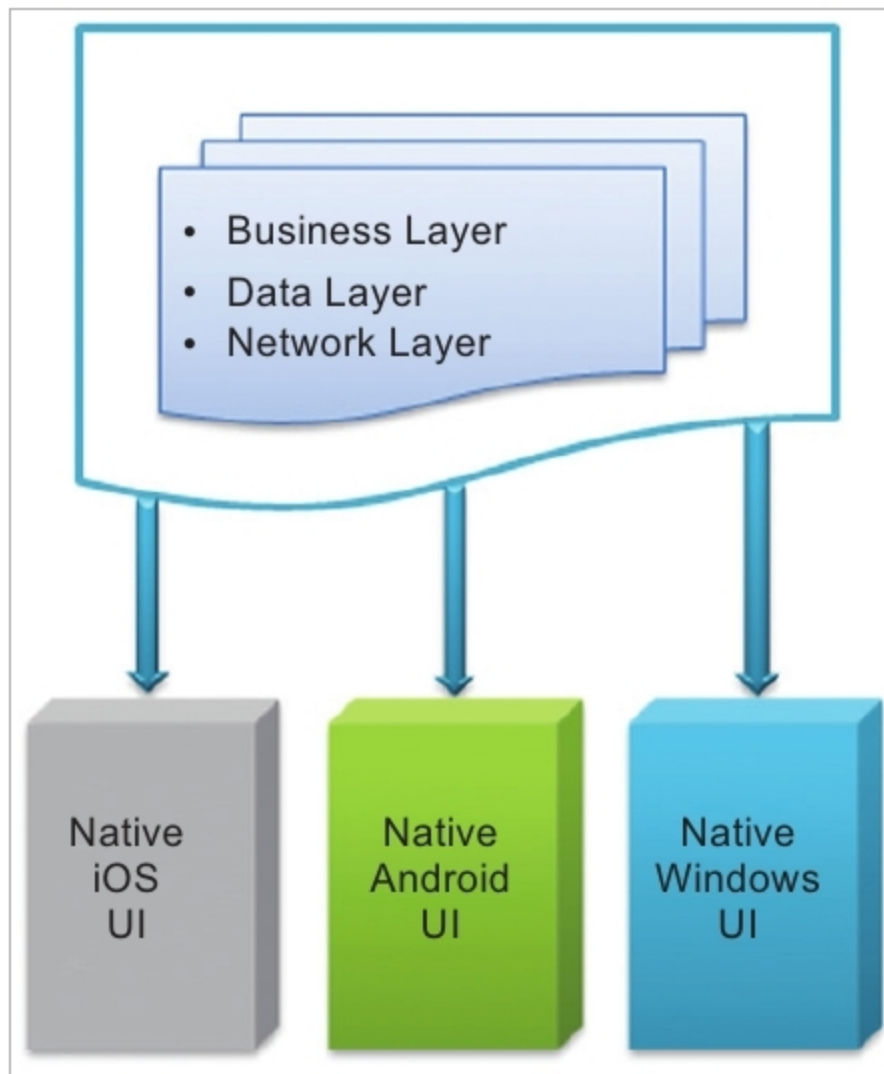


Figure 1: The capabilities of the underlying platform

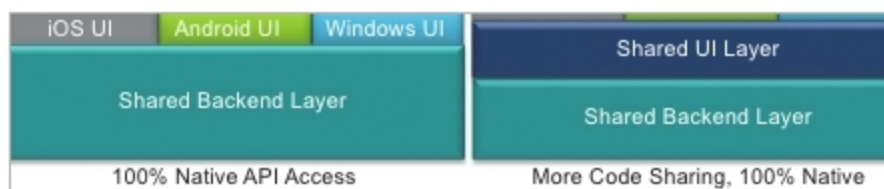


Figure 2: Xamarin Native and Xamarin Forms

The following are the system requirements to install Xamarin on MacOS and Windows. The IDE is Visual Studio for Mac and for Windows.

The macOS requirements are as follows.

Operating system: macOS Sierra 10.12 or above.

Xamarin.Android: Android 6.0 / API level 23. While using the latest SDK, you can still target older Android versions.

The Windows requirements are as follows.

Operating system: Recommended Windows 10. The minimum OS version is Windows 7.

Xamarin.Android: Android 6.0 / API level 23. While using the latest SDK, you can still target older Android versions.

Note: You can use any version of Visual Studio 2017—community, professional, and enterprise.

To install Xamarin on Windows and Mac, download the latest version of Visual Studio for the given OS and follow the installation instructions. For additional information on Xamarin installation, visit <https://docs.microsoft.com/en-us/xamarin/cross-platform/get-started/installation/index?tabs=windows>.

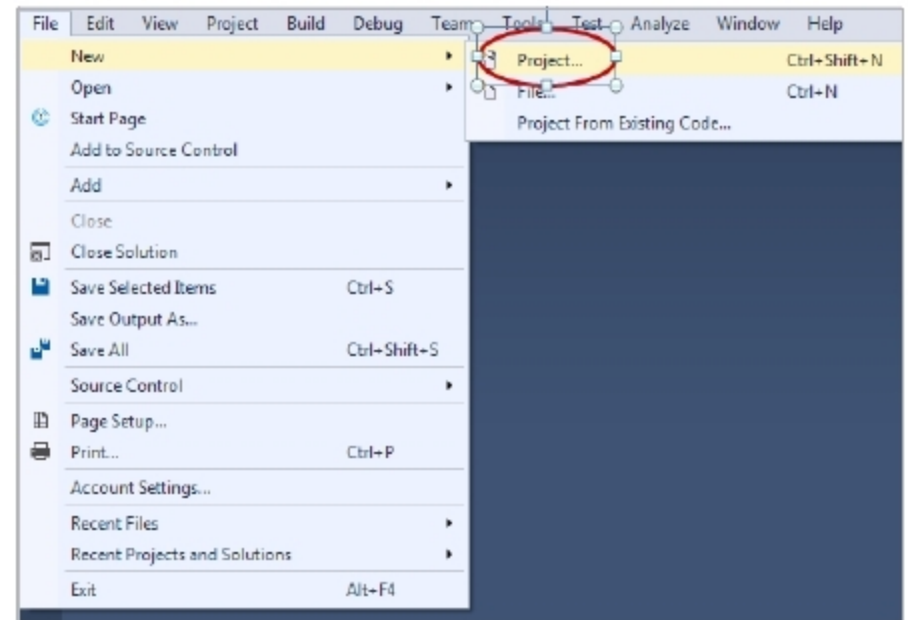


Figure 3: Microsoft Visual Studio – Creating a new project

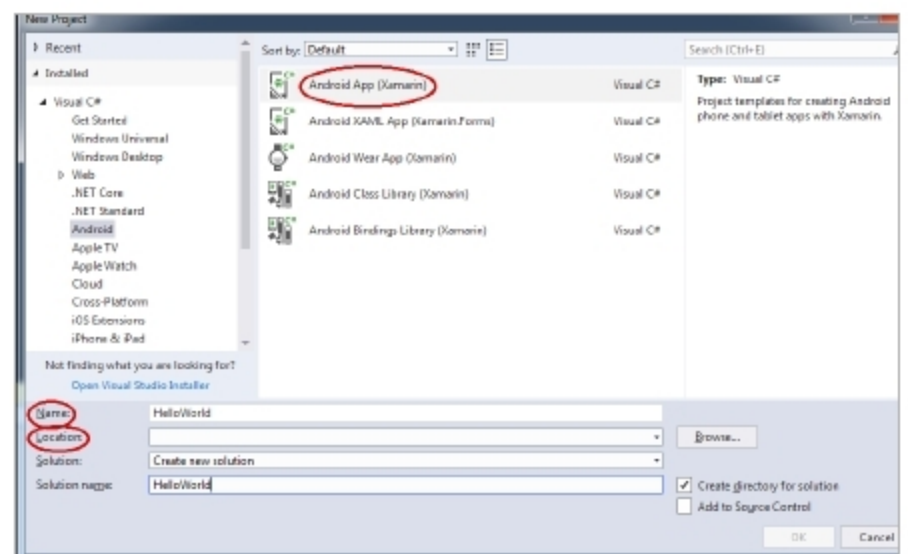


Figure 4: Creating an Android app in Microsoft Visual Studio using Xamarin

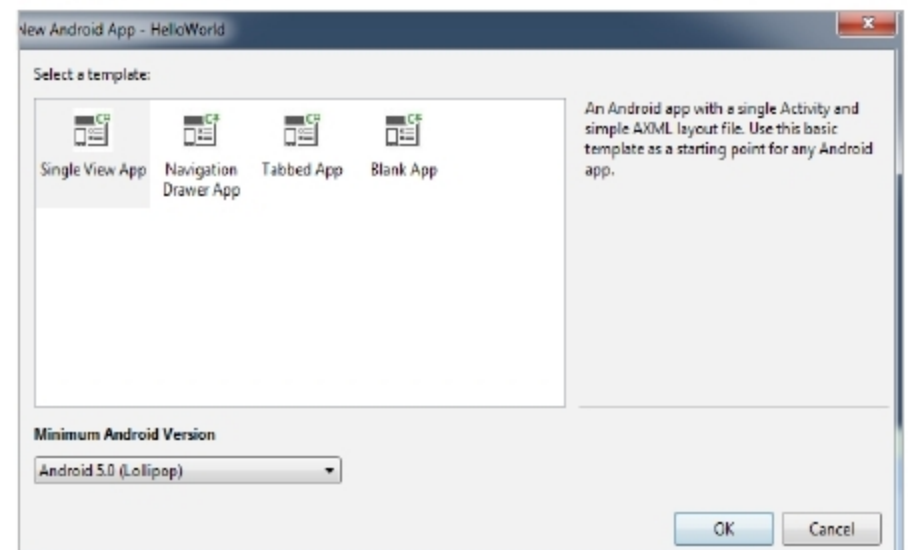


Figure 5: Single View App

Creating a sample Xamarin Android app

Open Visual Studio. Click on *File-> New-> Project*. This will bring up the *New Project* dialogue, as shown in Figure 4.

Next, select *Visual Studio-> Android-> Android App (Xamarin)*. You will need to fill out the following entries.

- **Application name:** Please type in *HelloWorld*.
- **Project location:** Select *Project Location*.

Next, tick the 'Create directory for solution' check box, and click on the OK button. This will bring up the next step, as shown in Figure 5.